

Learning to Rank — the reranker as a scoring function

20 years of building better estimators of relevance

QUERY × PROPERTY FEATURES

Booking.com property ranking

price

€89/night

review score

8.7 / 10

star rating

★★★★☆ 4 stars

location

0.8 km city centre

availability

12 rooms left

images

14 high-quality

reviews

847 reviews

description

complete + recent

The reranker is not magic, it has better features. The choice is cost vs. quality.

RE-RANKER

choose depth based on cost and quality

LTR

LambdaMART · GBDT

feature-weighted · fast

BERT cross-encoder

fill doc × query attention

better features · slower

LLM re-ranker

GPT-4 · Claude

human-level comprehension

most accurate · most expensive

RANKED RESULTS

sorted by learned score

1

Conservatorium Hotel

5★ · Score: 0.94

Museumplein · €285 · 9.2/10

248 nights booked

2

Hotel V Nesplein

4★ · Score: 0.87

City Centre · €149 · 8.8

3

The Student Hotel

3★ · Score: 0.79

De Pijp · €119 · 8.4